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Institute of electrical and optical communications

Advanced optical fiber communications

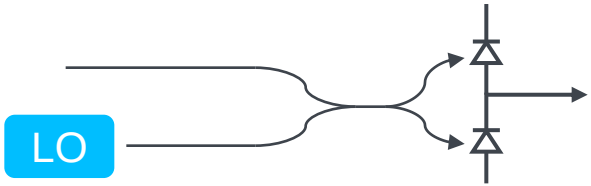
Wintersemester 2025/26

Lecture 11



Prof. Dr.-Ing. Georg Rademacher

Coherent Detection

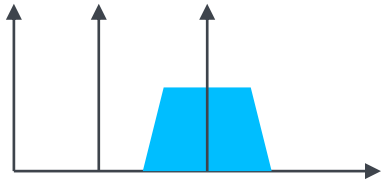


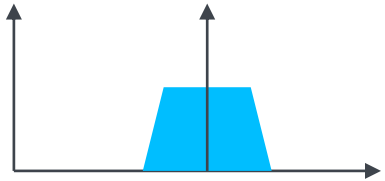
Coherent Detection

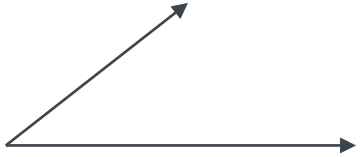
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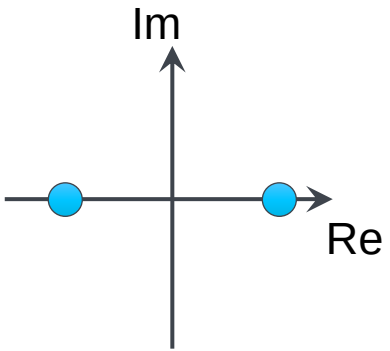
$$I(t) =$$

Homodyne Receivers

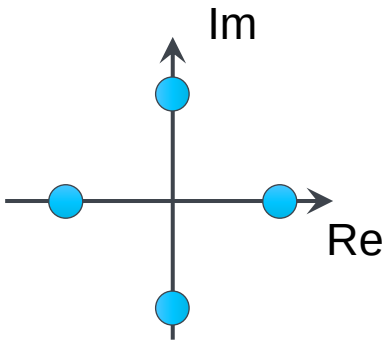


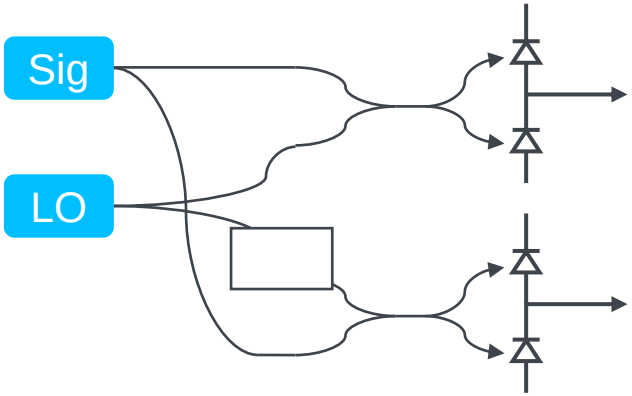






Homodyne Receivers: I and Q





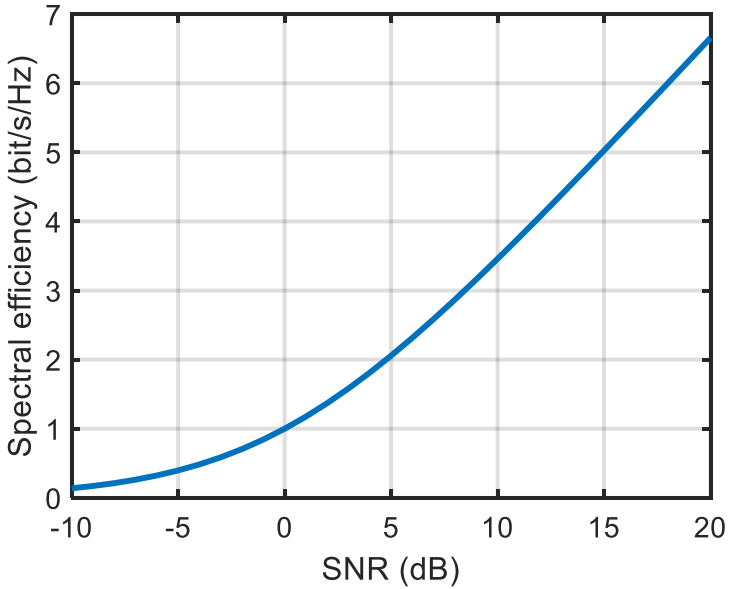
JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 34, NO. 1, JANUARY 1, 2016

Fundamentals of Coherent Optical Fiber Communications

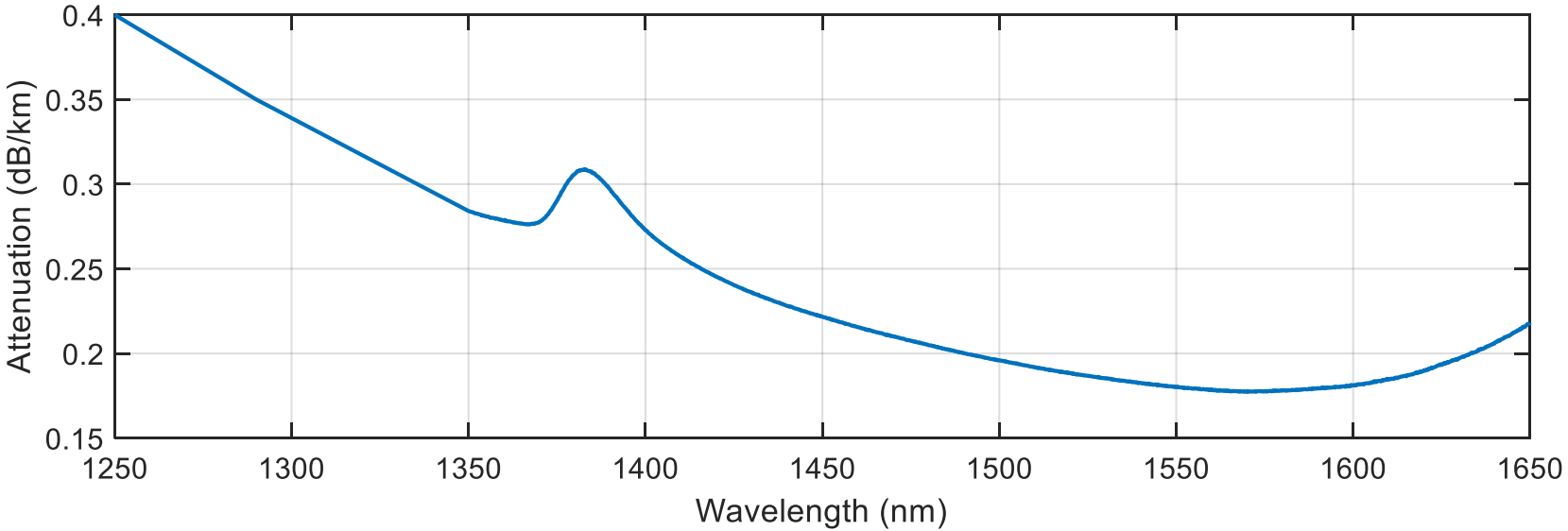
Kazuro Kikuchi, *Fellow, IEEE, Fellow, OSA*

(Tutorial Review)

Optical transmission link

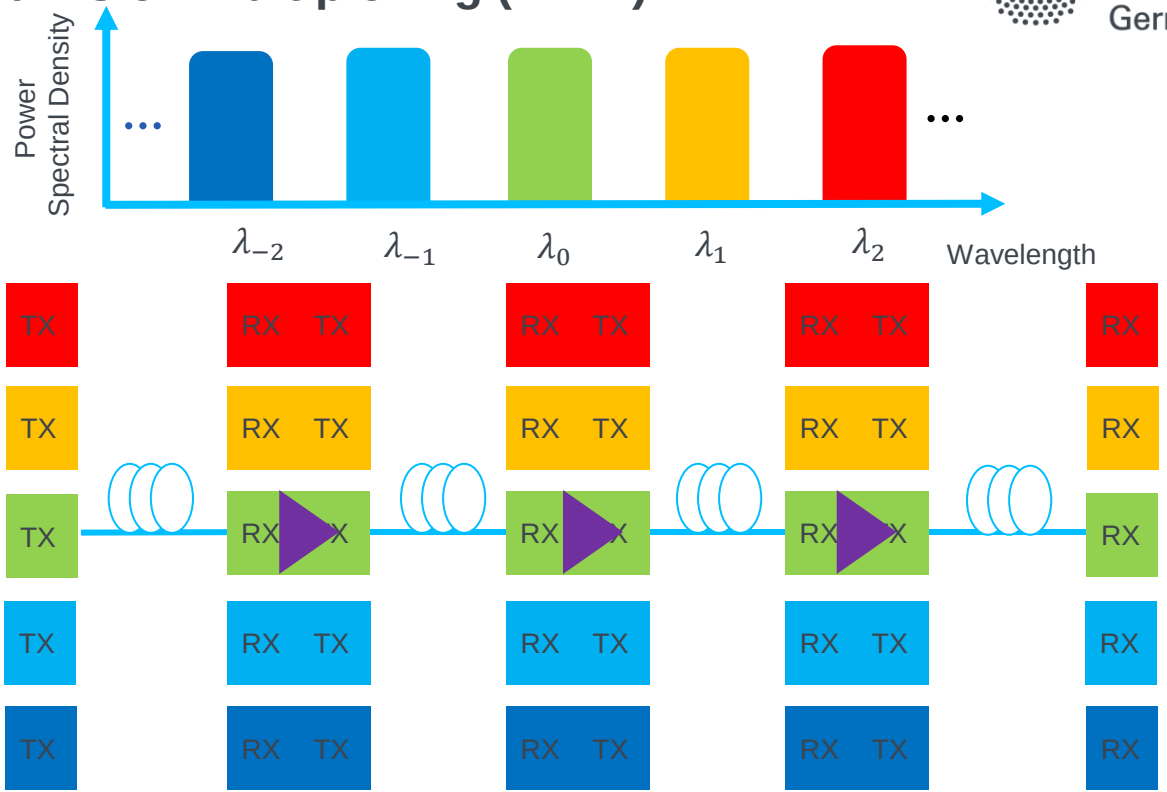


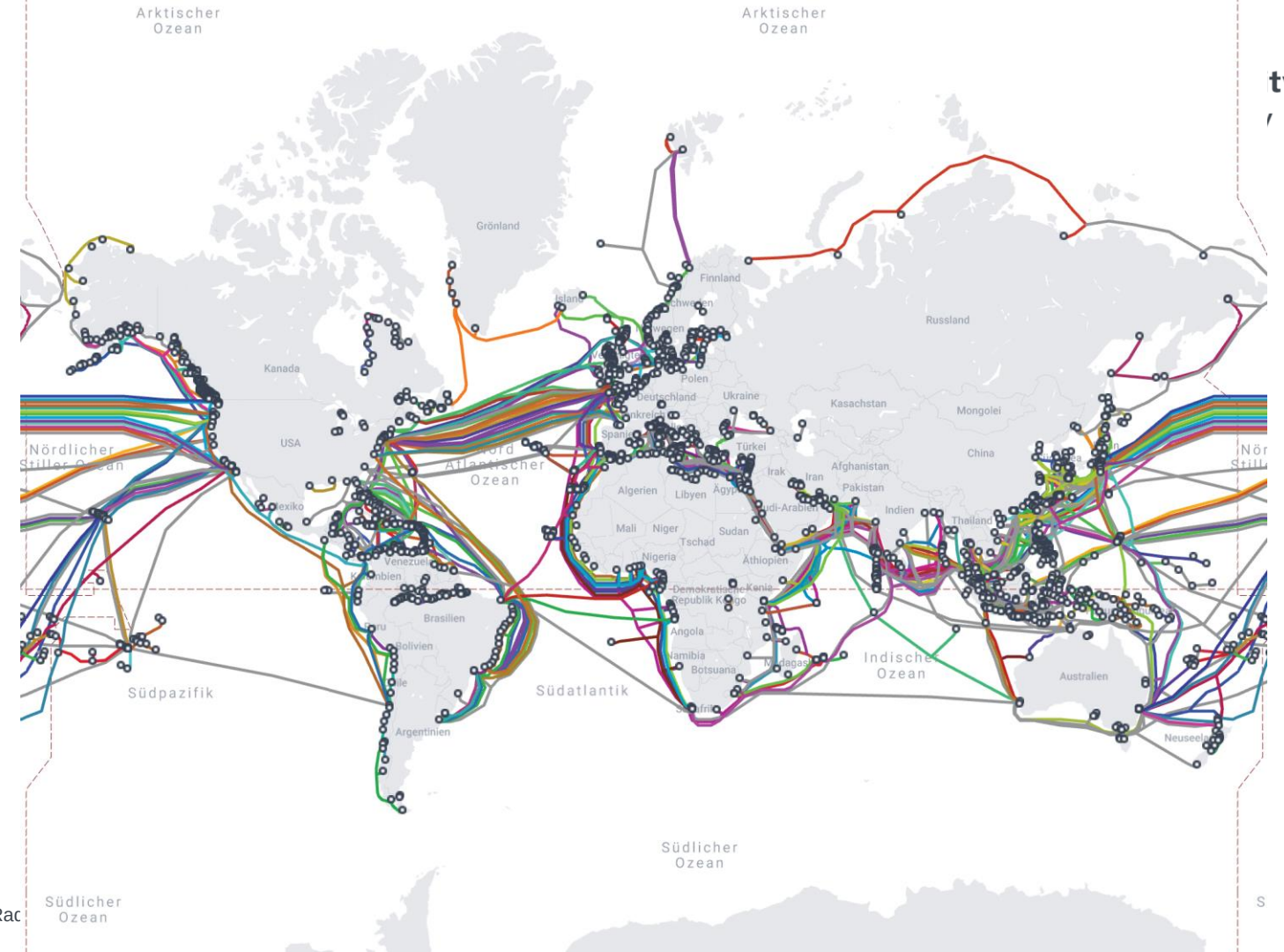
Wavelength-division multiplex (WDM)



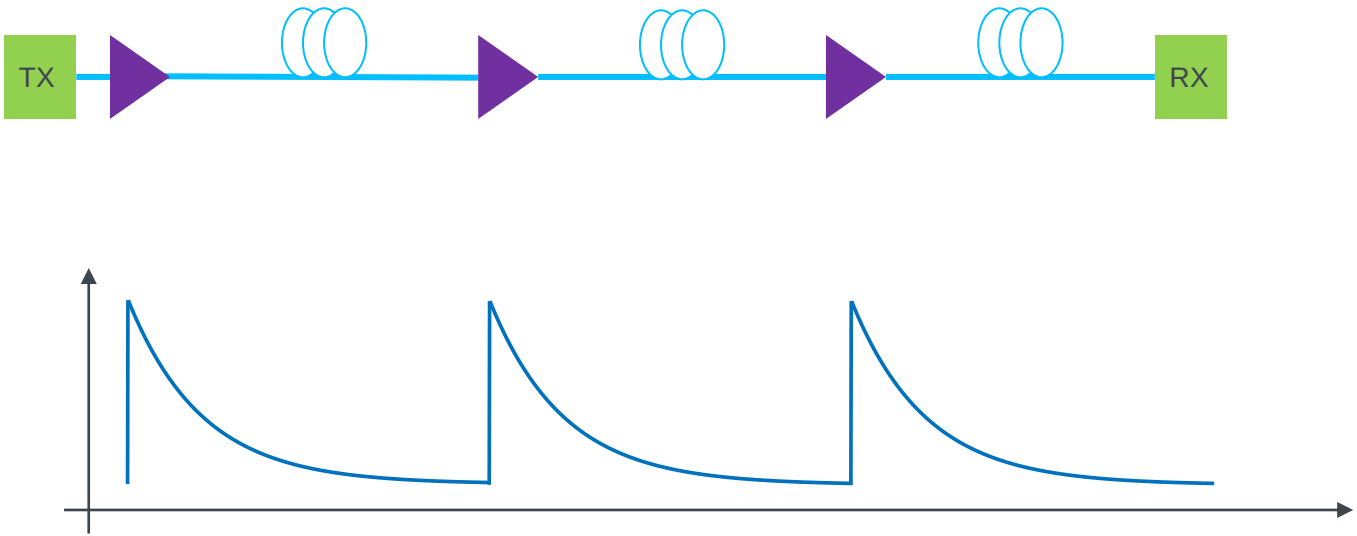
Wavelength division multiplexing (WDM)

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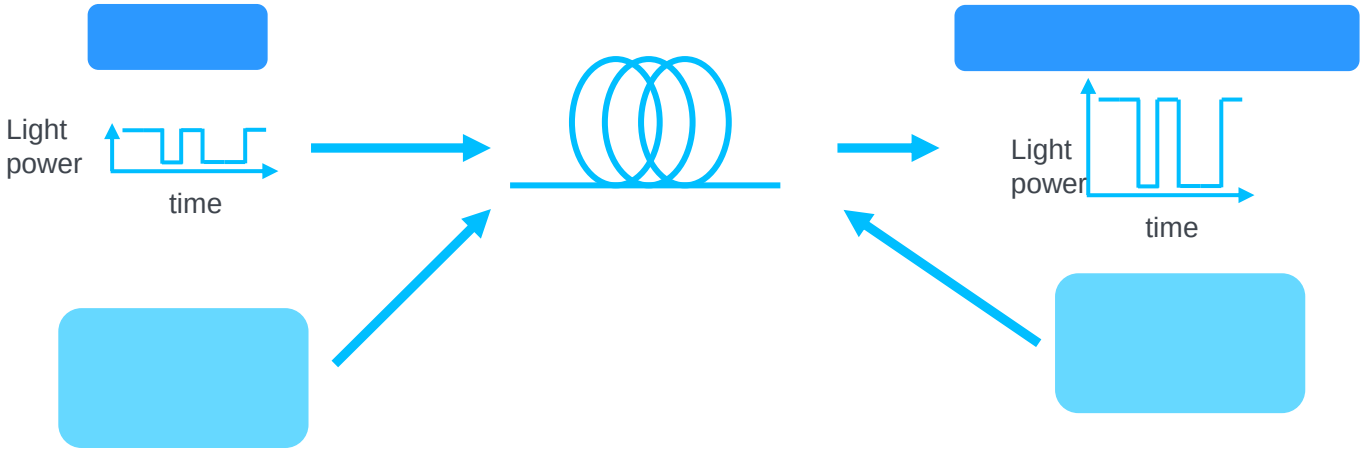

Arktischer Ozean
Arktischer Ozean
Grönland
Finnland
Russland
Kanada
Polen
Deutschland
Ukraine
Kasachstan
Mongolei
USA
Spanien
Türkei
China
Nördlicher Stiller Ozean
Nord Atlantischer Ozean
Irak
Iran
Afghanistan
Pakistan
Algerien
Libyen
Ägypten
Indien
Thailand
Mexiko
Mali
Niger
Sudan
Tschad
Nigeria
Äthiopien
Kenia
Demokratische Republik Kongo
Venezuela
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Australien
Neuseeland
Argentinien
Südlicher Ozean
Südlicher Ozean

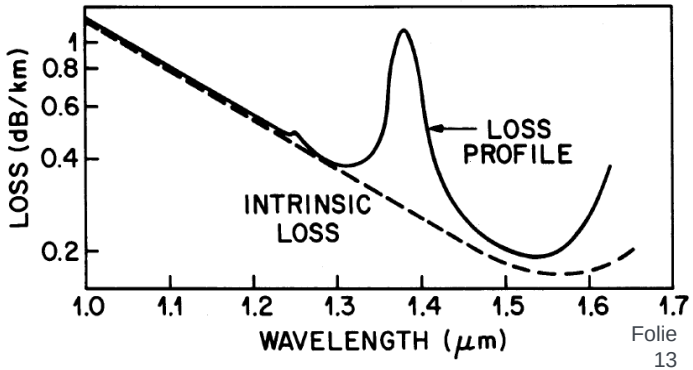
Power profile in amplified transmission link



Erbium doped fiber amplifier (EDFA)

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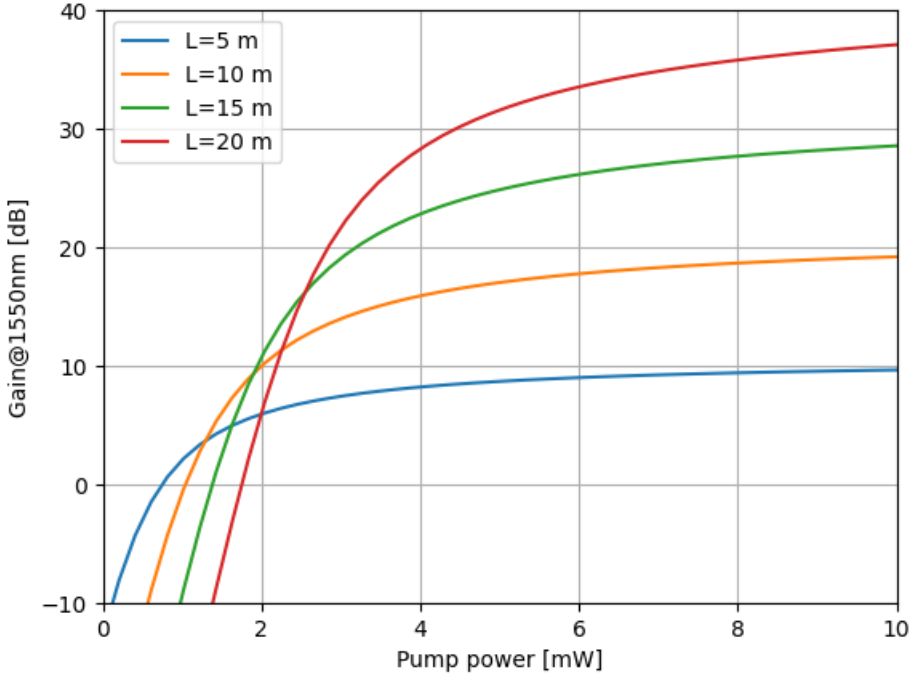
Erbium-doped fiber amplifier (EDFA)

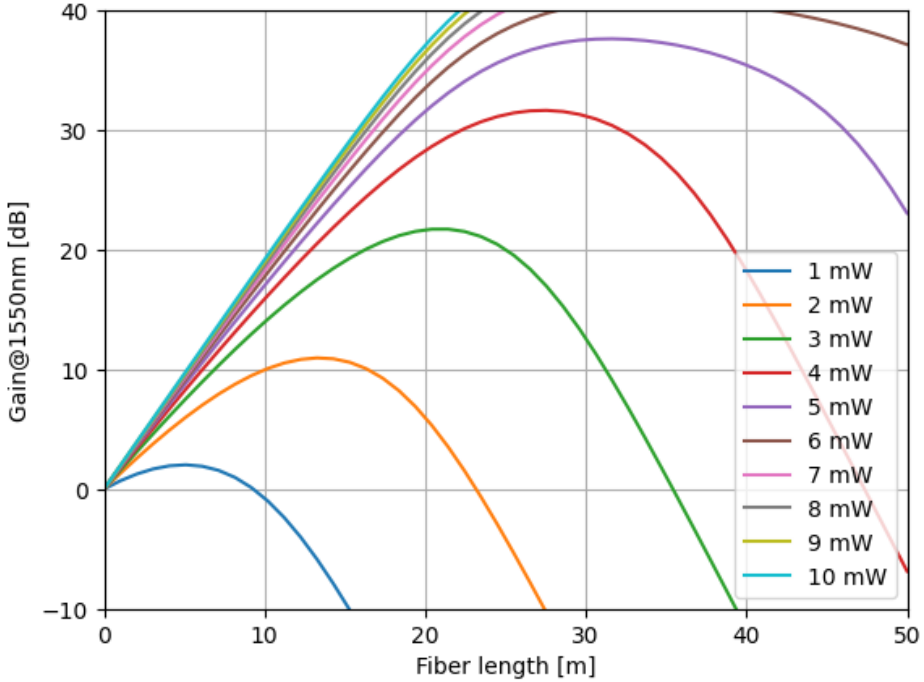


Pump and signal power evolution

Gain vs. pump power and fiber length

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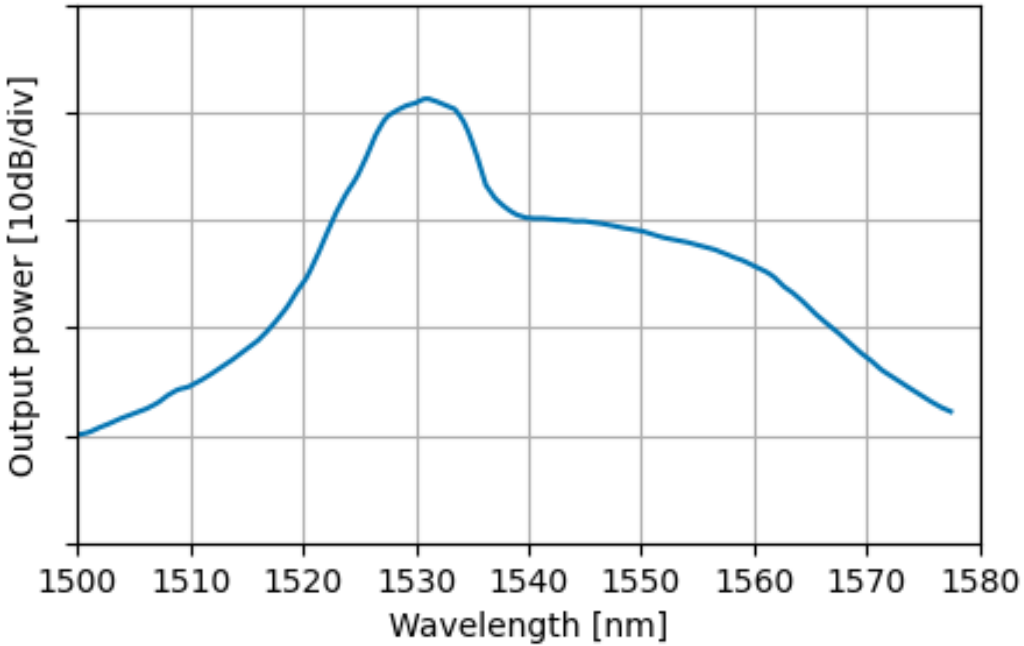




EDFA Noise

EDFA Noise

EDFA Noise


Output power [10dB/div]
1500
1510
1520
1530
1540
1550
1560
1570
1580
Wavelength [nm]

Noise in long distance transmission

2. Friis formula for two blocks

of Stuttgart

For **two cascaded stages**:



Noise in long distance transmission

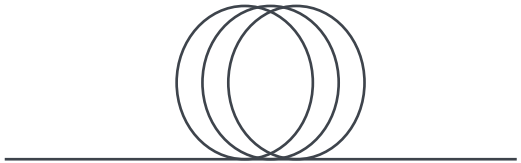
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Faserverstärker mit der Rauschzahl NF_1

Koppelwirkungsgrad η und damit einer Rauschzahl $1/\eta$ be

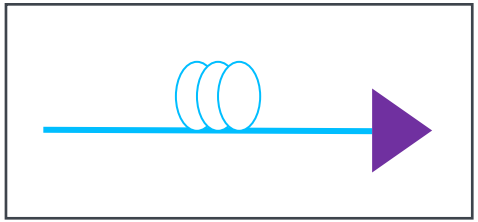
$$NF_1 = 2n_{sp}\left(1 - \frac{1}{G}\right) + \frac{1}{G}$$

$$\overline{NF_2} = \frac{1}{G_F}$$

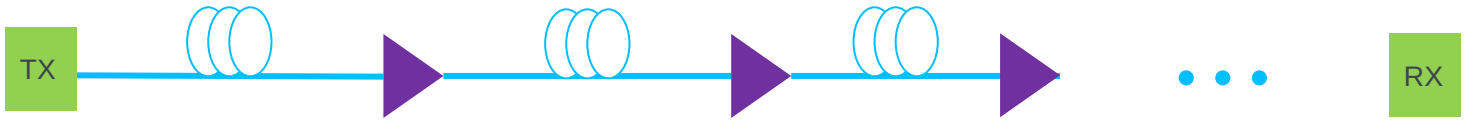


$$NF_{total} = \frac{1}{\eta} + \frac{NF_1 - 1}{\eta} = \frac{NF_1}{\eta}$$

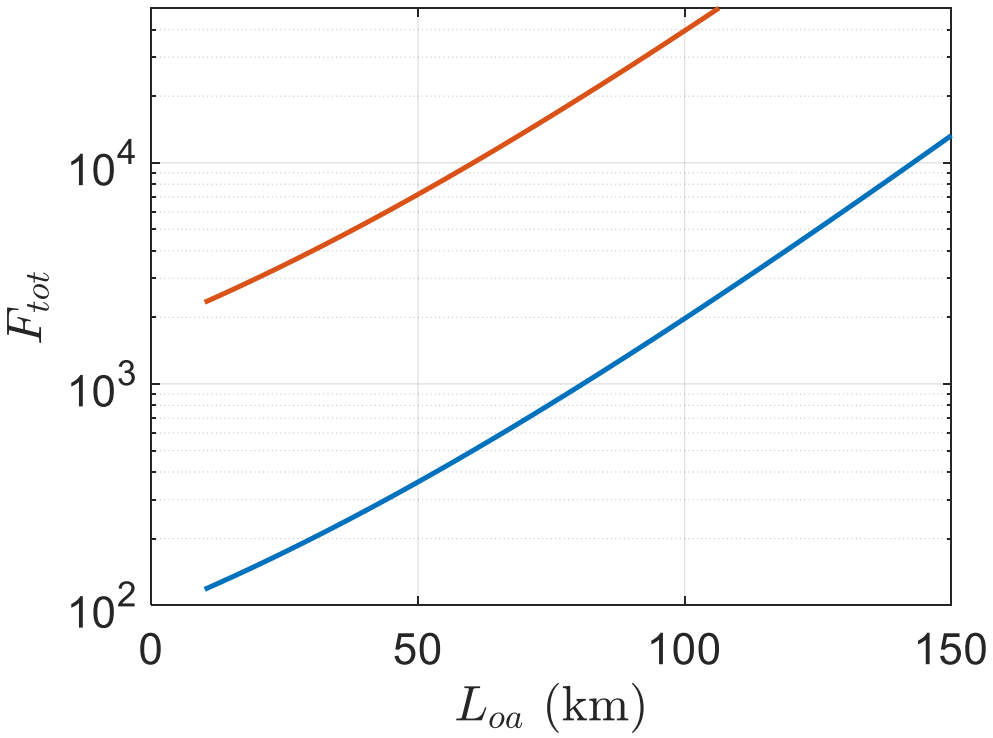
Noise in long distance transmission



Noise in long distance transmission



Noise in long distance transmission

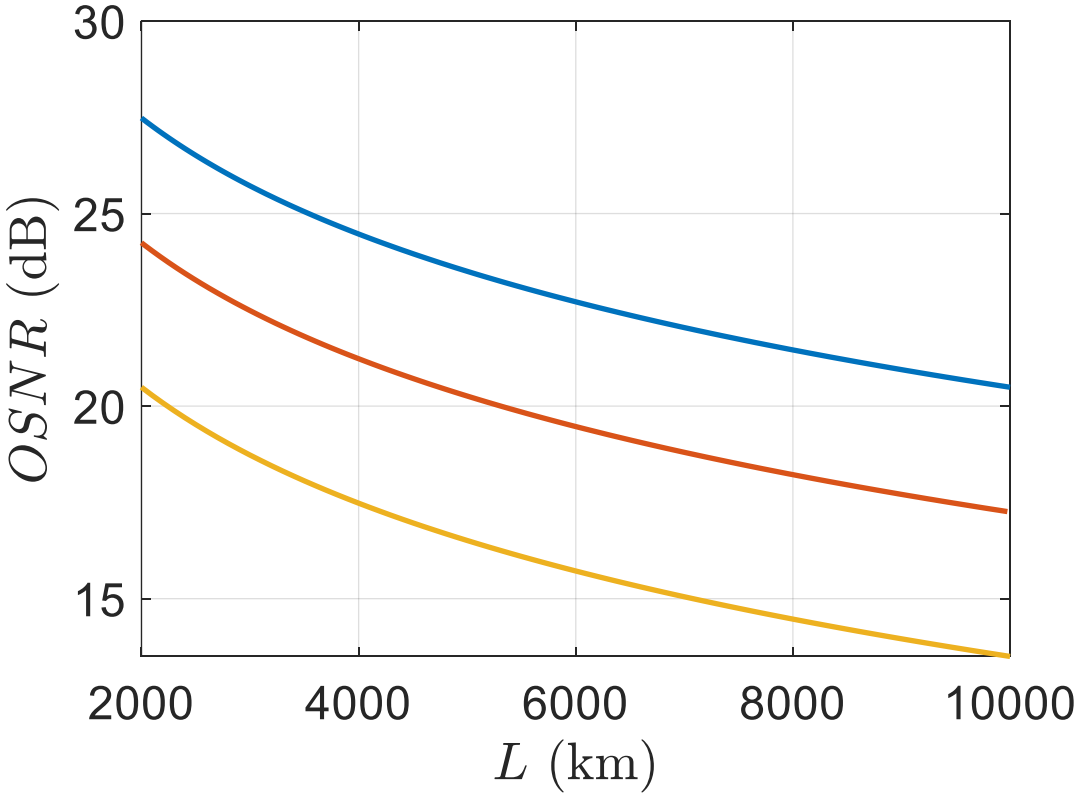


Optical Signal to Noise Ratio (OSNR)

OSNR in long distance transmission

OSNR in long distance transmission

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Thank you!

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